RETAIL SALES PERFORMANCE ANALYSIS

1.Project Overview and Objective:

- The Retail Sales Performance Analysis project focuses on analysing sales transactions, customer details, product categories, payment methods, and sales channels. The dataset contains 500 records and 16 columns, including order details, customer information, product details, quantity, unit price, discount percentage, sales amount, payment method, order status, sales channel, and revenue.

- This project uses Microsoft Excel features such as data cleaning, formulas, Pivot Tables, and data analysis to evaluate sales performance, identify popular product categories, compare sales across different cities, and understand customer purchasing patterns. The analysis aims to transform raw retail sales data into meaningful insights that support effective business decision-making.

2.Data Sources:

- Source Description and Timeline: Retail Sales Dataset used for this project, containing 500 sales transaction records. https://www.zoho.com/analytics/sample-data-generator.html

- Domain: Retail / Sales Analytics.

3.Problem Statement:

- Retail businesses generate large amounts of sales data involving different customers, products, cities, payment methods, and sales channels.

- However, raw sales data alone does not provide a clear understanding of business performance, customer purchasing behaviour, or product sales trends.
- The challenge is to analyse the available retail sales dataset to identify which product categories generate higher revenue, which cities contribute more to sales, and which payment methods and sales channels are commonly used.
- By using Excel-based data cleaning, formulas, Pivot Tables, and visualization techniques, this project aims to identify meaningful patterns, uncover business insights, and support informed decisions to improve retail sales performance.

4.Attribute (Column / Features) Details:

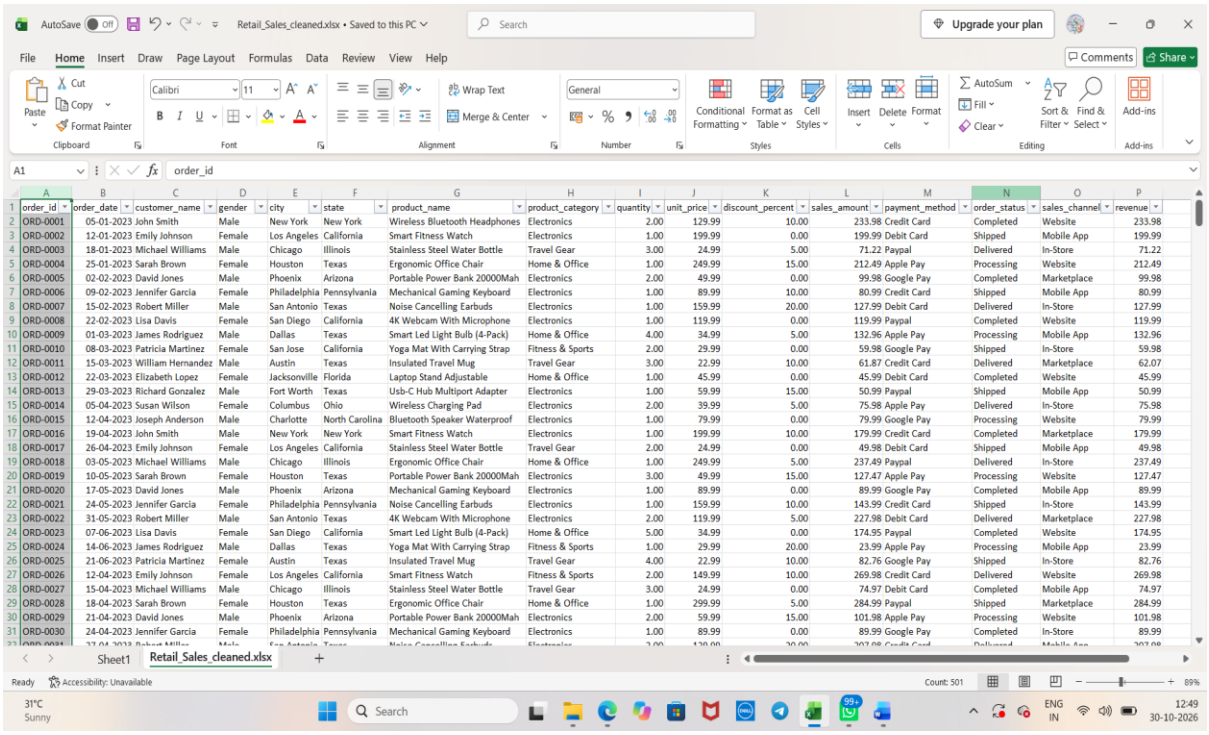
Attribute Name	Data Type	Description
Order ID	Text	Unique identification number assigned to each order.
Order Date	Date	Date when the customer placed the order.
Customer Name	Text	Name of the customer who placed the order.
Gender	Text	Gender category associated with the customer.
City	Text	City where the customer is located.
State	Text	State where the customer is located.
Product Name	Text	Name of the product purchased.
Product Category	Text	Category or type of product sold.
Quantity	Whole Number	Total number of units purchased in an order.
Unit Price	Decimal	Selling price of a single product unit.
Discount Percent	Percentage	Discount percentage applied to the order.
Sales Amount	Decimal / Currency	Sales amount generated from the transaction.
Payment Method	Text	Method used by the customer to make payment.
Order Status	Text	Current status of the order, such as Completed, Shipped, Delivered, or Processing.
Sales Channel	Text	Platform or channel through which the customer placed the order.
Revenue	Decimal / Currency	Revenue generated from the sales transaction.

5.Tools & Technologies:

- Excel: Data cleaning, transformation, and Pivot Tables.
- Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6.Data Pre-Processing :

➢ Data Cleaning And Transformation:



order_id	order_date	customer_name	gender	city	state	product_name	product_category	quantity	unit_price	discount_percent	sales_amount	payment_method	order_status	sales_channel	revenue
ORD-0001	05-01-2023	John Smith	Male	New York	New York	Wireless Bluetooth Headphones	Electronics	2.00	129.99	10.00	233.98	Credit Card	Completed	Website	233.98
ORD-0002	12-01-2023	Emily Johnson	Female	Los Angeles	California	Smart Fitness Watch	Electronics	1.00	199.99	0.00	199.99	Debit Card	Shipped	Mobile App	199.99
ORD-0003	18-01-2023	Michael Williams	Male	Chicago	Illinois	Stainless Steel Water Bottle	Travel Gear	3.00	24.99	5.00	71.22	Paypal	Delivered	In-Store	71.22
ORD-0004	25-01-2023	Sarah Brown	Female	Houston	Texas	Ergonomic Office Chair	Home & Office	1.00	249.99	15.00	212.49	Apple Pay	Processing	Website	212.49
ORD-0005	02-02-2023	David Jones	Male	Phoenix	Arizona	Portable Power Bank 20000Mah	Electronics	2.00	49.99	0.00	99.98	Google Pay	Completed	Marketplace	99.98
ORD-0006	09-02-2023	Jennifer Garcia	Female	Philadelphia	Pennsylvania	Mechanical Gaming Keyboard	Electronics	1.00	89.99	10.00	80.99	Credit Card	Shipped	Mobile App	80.99
ORD-0007	15-02-2023	Robert Miller	Male	San Antonio	Texas	Noise Cancelling Earbuds	Electronics	1.00	159.99	20.00	127.99	Debit Card	Delivered	In-Store	127.99
ORD-0008	22-02-2023	Lisa Davis	Female	San Diego	California	4K Webcam With Microphone	Electronics	1.00	119.99	0.00	119.99	Paypal	Completed	Website	119.99
ORD-0009	01-03-2023	James Rodriguez	Male	Dallas	Texas	Smart Led Light Bulb (4-Pack)	Home & Office	4.00	34.99	5.00	132.96	Apple Pay	Processing	Mobile App	132.96
ORD-0010	08-03-2023	Patricia Martinez	Female	San Jose	California	Yoga Mat With Carrying Strap	Fitness & Sports	2.00	29.99	0.00	59.98	Google Pay	Shipped	In-Store	59.98
ORD-0011	15-03-2023	William Hernandez	Male	Austin	Texas	Insulated Travel Mug	Travel Gear	3.00	22.99	10.00	61.87	Credit Card	Delivered	Marketplace	62.07
ORD-0012	22-03-2023	Elizabeth Lopez	Female	Jacksonville	Florida	Laptop Stand Adjustable	Home & Office	1.00	45.99	0.00	45.99	Debit Card	Completed	Website	45.99
ORD-0013	29-03-2023	Richard Gonzalez	Male	Fort Worth	Texas	Usb-C Hub Multiport Adapter	Electronics	1.00	59.99	15.00	50.99	Paypal	Shipped	Mobile App	50.99
ORD-0014	05-04-2023	Susan Wilson	Female	Columbus	Ohio	Wireless Charging Pad	Electronics	2.00	39.99	5.00	75.98	Apple Pay	Delivered	In-Store	75.98
ORD-0015	12-04-2023	Joseph Anderson	Male	Charlotte	North Carolina	Bluetooth Speaker Waterproof	Electronics	1.00	79.99	0.00	79.99	Google Pay	Processing	Website	79.99
ORD-0016	19-04-2023	John Smith	Male	New York	New York	Smart Fitness Watch	Electronics	1.00	199.99	10.00	179.99	Credit Card	Completed	Marketplace	179.99
ORD-0017	26-04-2023	Emily Johnson	Female	Los Angeles	California	Stainless Steel Water Bottle	Travel Gear	2.00	24.99	0.00	49.98	Debit Card	Shipped	Mobile App	49.98
ORD-0018	03-05-2023	Michael Williams	Male	Chicago	Illinois	Ergonomic Office Chair	Home & Office	1.00	249.99	5.00	237.49	Paypal	Delivered	In-Store	237.49
ORD-0019	10-05-2023	Sarah Brown	Female	Houston	Texas	Portable Power Bank 20000Mah	Electronics	3.00	49.99	15.00	127.47	Apple Pay	Processing	Website	127.47
ORD-0020	17-05-2023	David Jones	Male	Phoenix	Arizona	Mechanical Gaming Keyboard	Electronics	1.00	89.99	0.00	89.99	Google Pay	Completed	Mobile App	89.99
ORD-0021	24-05-2023	Jennifer Garcia	Female	Philadelphia	Pennsylvania	Noise Cancelling Earbuds	Electronics	1.00	159.99	10.00	143.99	Credit Card	Shipped	In-Store	143.99
ORD-0022	31-05-2023	Robert Miller	Male	San Antonio	Texas	4K Webcam With Microphone	Electronics	2.00	119.99	5.00	227.98	Debit Card	Delivered	Marketplace	227.98
ORD-0023	07-06-2023	Lisa Davis	Female	San Diego	California	Smart Led Light Bulb (4-Pack)	Home & Office	5.00	34.99	0.00	174.95	Paypal	Completed	Website	174.95
ORD-0024	14-06-2023	James Rodriguez	Male	Dallas	Texas	Yoga Mat With Carrying Strap	Fitness & Sports	1.00	29.99	20.00	23.99	Apple Pay	Processing	Mobile App	23.99
ORD-0025	21-06-2023	Patricia Martinez	Female	Austin	Texas	Insulated Travel Mug	Travel Gear	4.00	22.99	10.00	82.76	Google Pay	Shipped	In-Store	82.76
ORD-0026	12-04-2023	Emily Johnson	Female	Los Angeles	California	Smart Fitness Watch	Fitness & Sports	2.00	149.99	10.00	269.98	Credit Card	Delivered	Website	269.98
ORD-0027	15-04-2023	Michael Williams	Male	Chicago	Illinois	Stainless Steel Water Bottle	Travel Gear	3.00	24.99	0.00	74.97	Debit Card	Completed	Mobile App	74.97
ORD-0028	18-04-2023	Sarah Brown	Female	Houston	Texas	Ergonomic Office Chair	Home & Office	1.00	299.99	5.00	284.99	Paypal	Shipped	Marketplace	284.99
ORD-0029	21-04-2023	David Jones	Male	Phoenix	Arizona	Portable Power Bank 20000Mah	Electronics	2.00	59.99	15.00	101.98	Apple Pay	Processing	Website	101.98
ORD-0030	24-04-2023	Jennifer Garcia	Female	Philadelphia	Pennsylvania	Mechanical Gaming Keyboard	Electronics	1.00	89.99	0.00	89.99	Google Pay	Completed	In-Store	89.99

➢ Formula Used:

❖ PROPER, TRIM and CLEAN

Formula:

"=PROPER(TRIM(CLEAN(C2)))"

Purpose:

- ❖ CLEAN removes non-printing characters.
- ❖ TRIM removes unwanted extra spaces.
- ❖ PROPER converts names into proper capitalization.
- ❖ Used for cleaning and standardizing customer names.

- ❖ **2. ROUND**

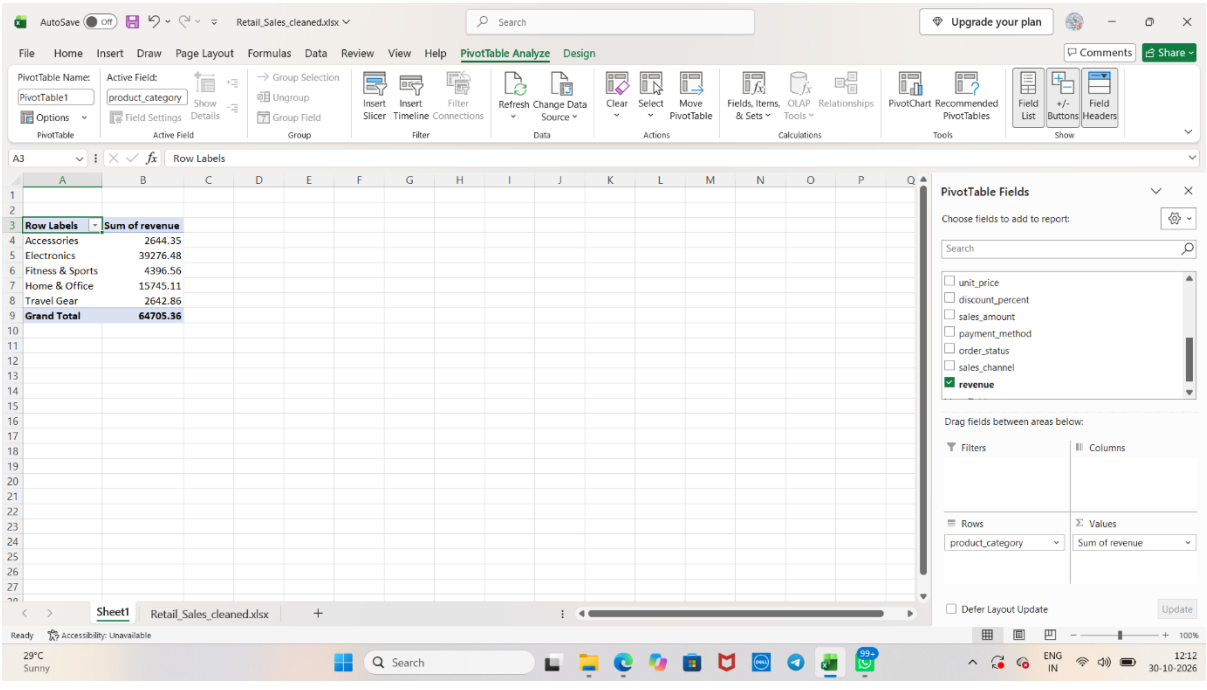
Formula:
"=ROUND(I2*J2*(1-K2/100),2)"

Purpose:

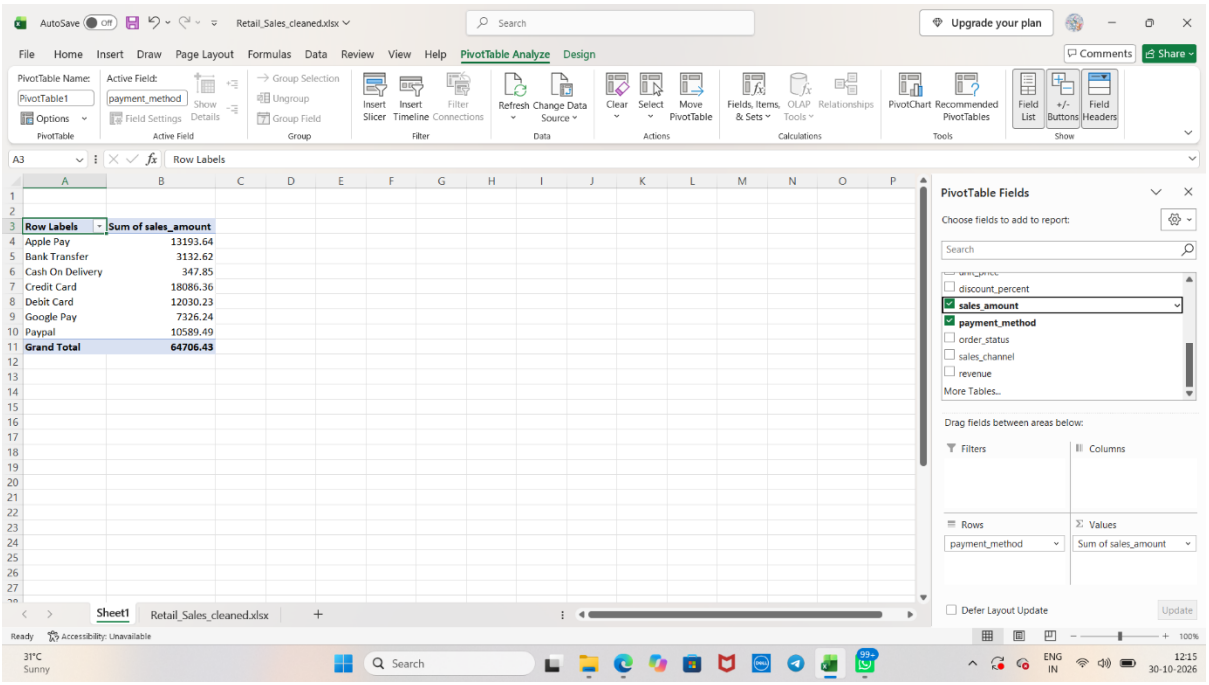
- ❖ Calculates revenue using Quantity, Unit Price, and Discount Percentage.
- ❖ Applies the discount to the product amount.
- ❖ ROUND ensures the final result contains two decimal places.

7.Pivot Table:

➢ Product Category-Wise Sales Analysis:

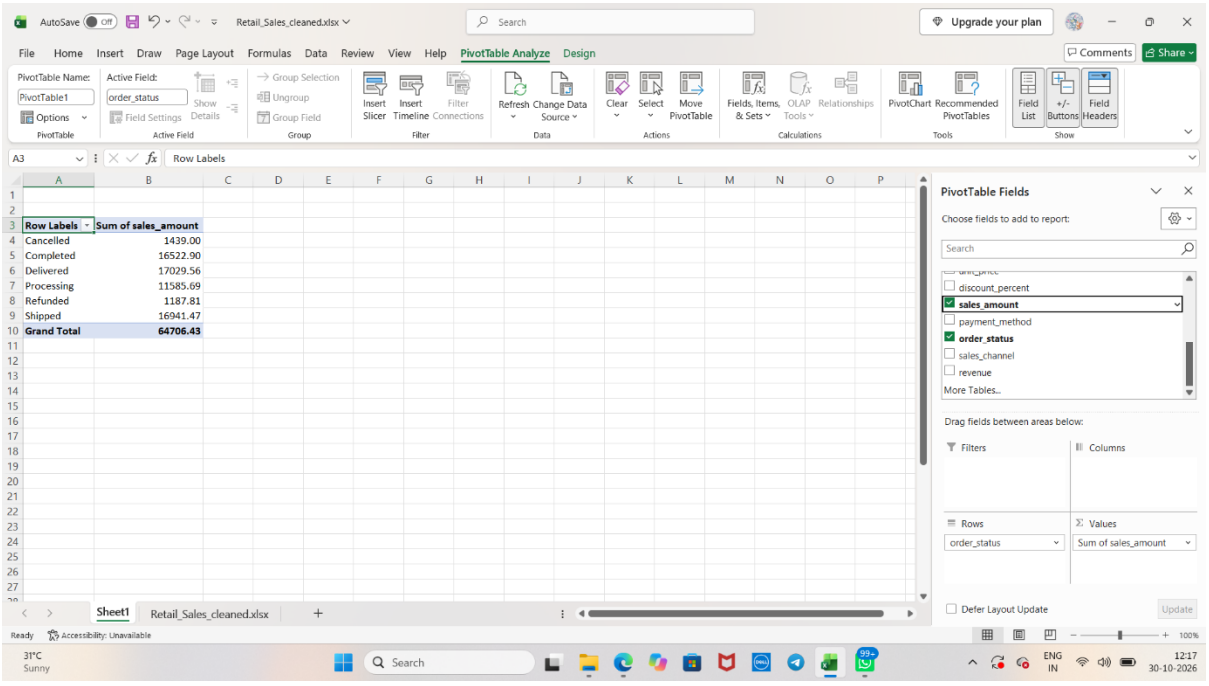


➢ Payment Method-Wise Sales Analysis:



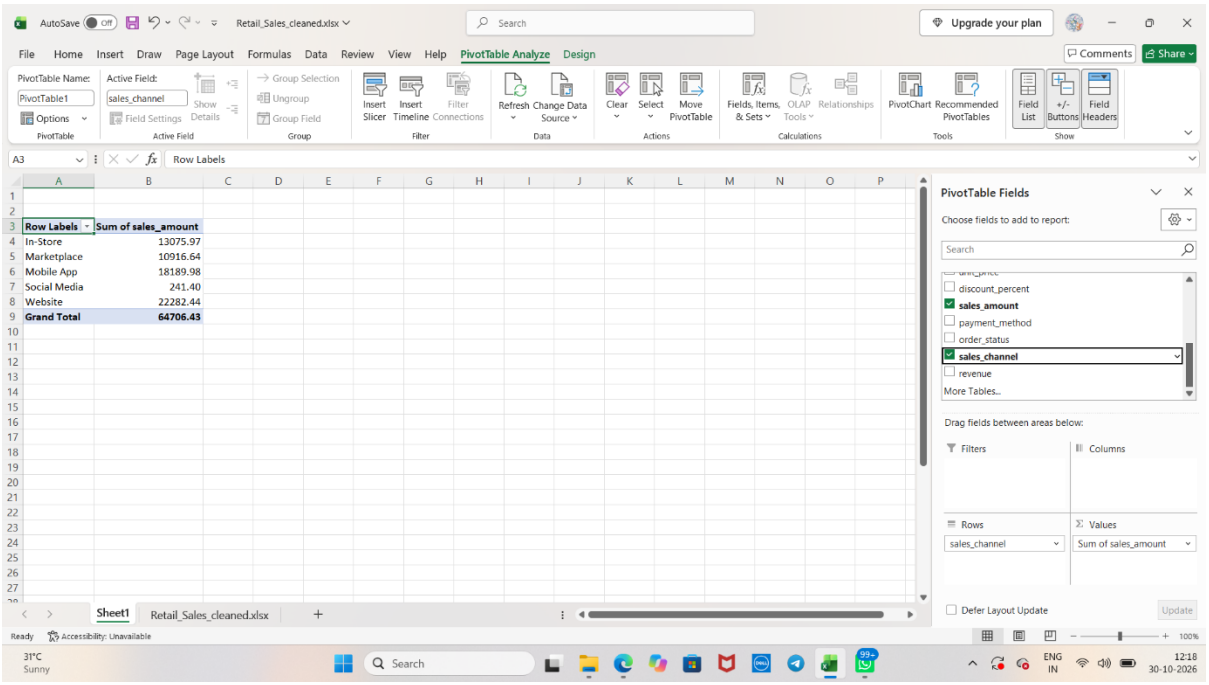
Row Labels	Sum of sales_amount
Apple Pay	13193.64
Bank Transfer	3132.62
Cash On Delivery	347.85
Credit Card	18086.36
Debit Card	12030.23
Google Pay	7326.24
Paypal	10589.49
Grand Total	**64706.43**

➢ Order Status-Wise Sales Analysis:

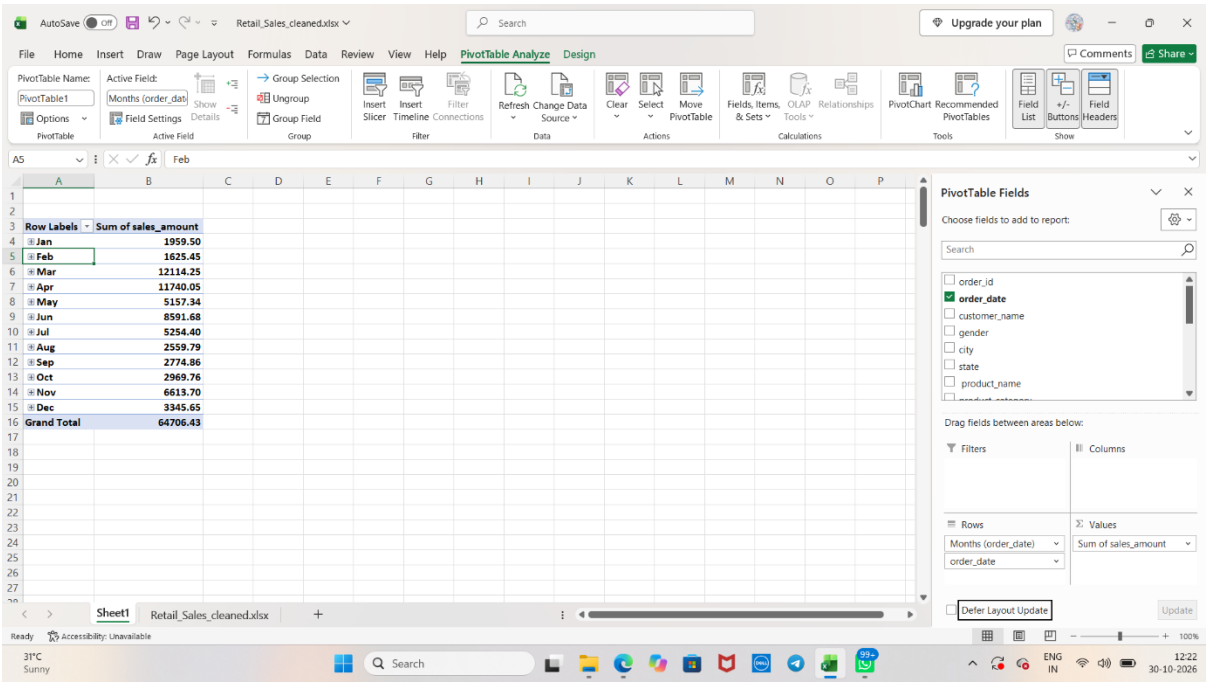


Row Labels	Sum of sales_amount
Cancelled	1439.00
Completed	16522.90
Delivered	17029.56
Processing	11585.69
Refunded	1187.81
Shipped	16941.47
Grand Total	**64706.43**

➢ Sales Channel-Wise Analysis:



➢ Month Wise Sales Analysis:



8.Key Findings:

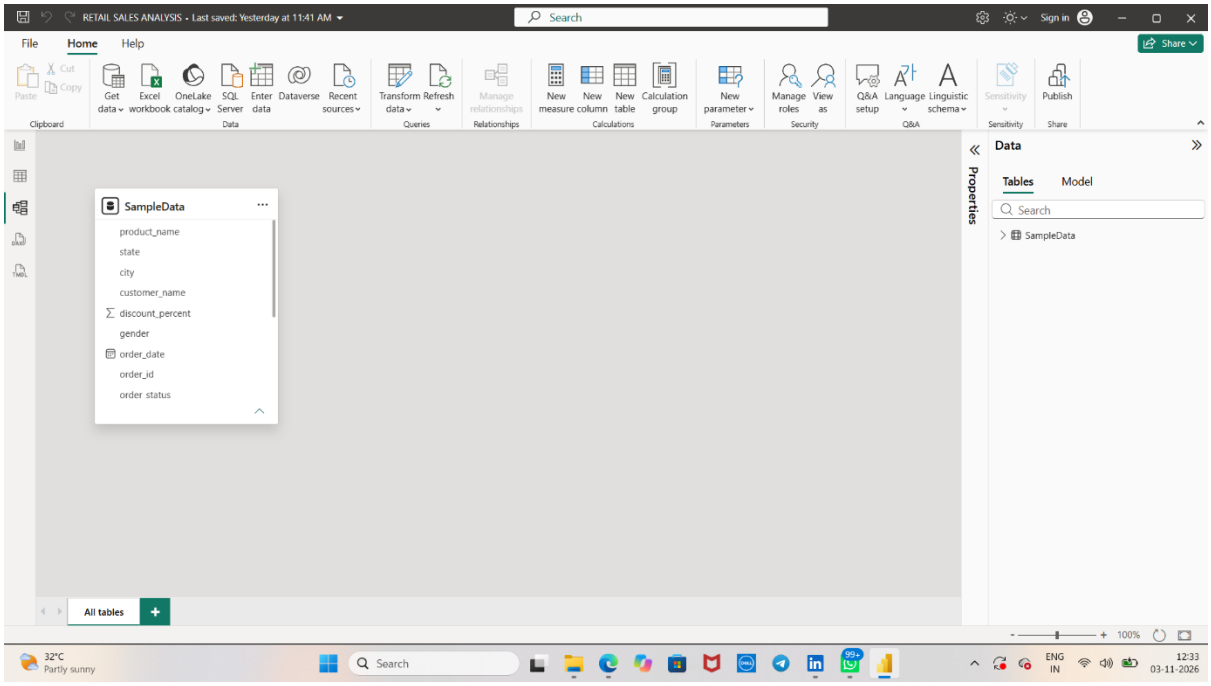
- Travel Gear recorded the lowest sales among product categories, with a total sales amount of 2,642.66.
- Website was the highest-performing sales channel, generating total sales of 22,282.44.
- Social Media recorded the lowest sales among sales channels, contributing 241.40.
- The total revenue generated from all retail transactions was 64,706.43.
- A total of 501 unique orders were recorded, with 934 products sold.
- Month-wise sales analysis helps identify variations in sales performance across different months.

9.Overall Observation:

The Pivot Table analysis provides a clear summary of retail sales performance and helps identify important patterns across months, payment methods, order statuses, sales channels, and product categories.

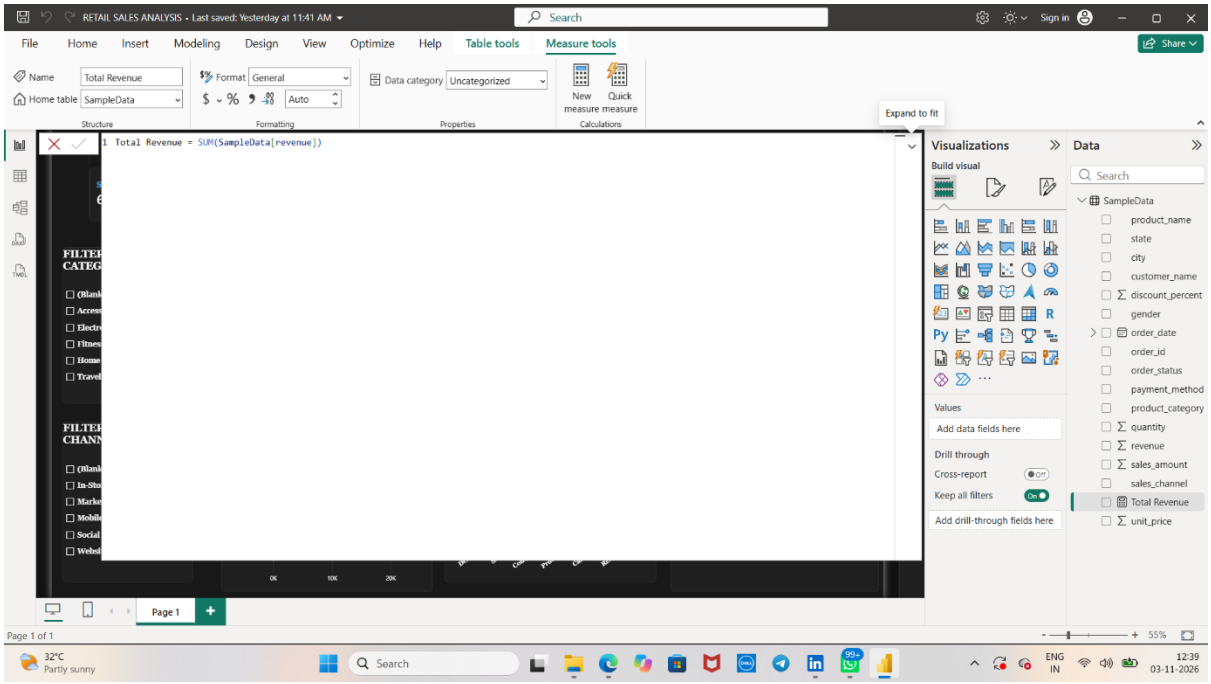
10.Data Modelling And DAX:

- **Data Model:**

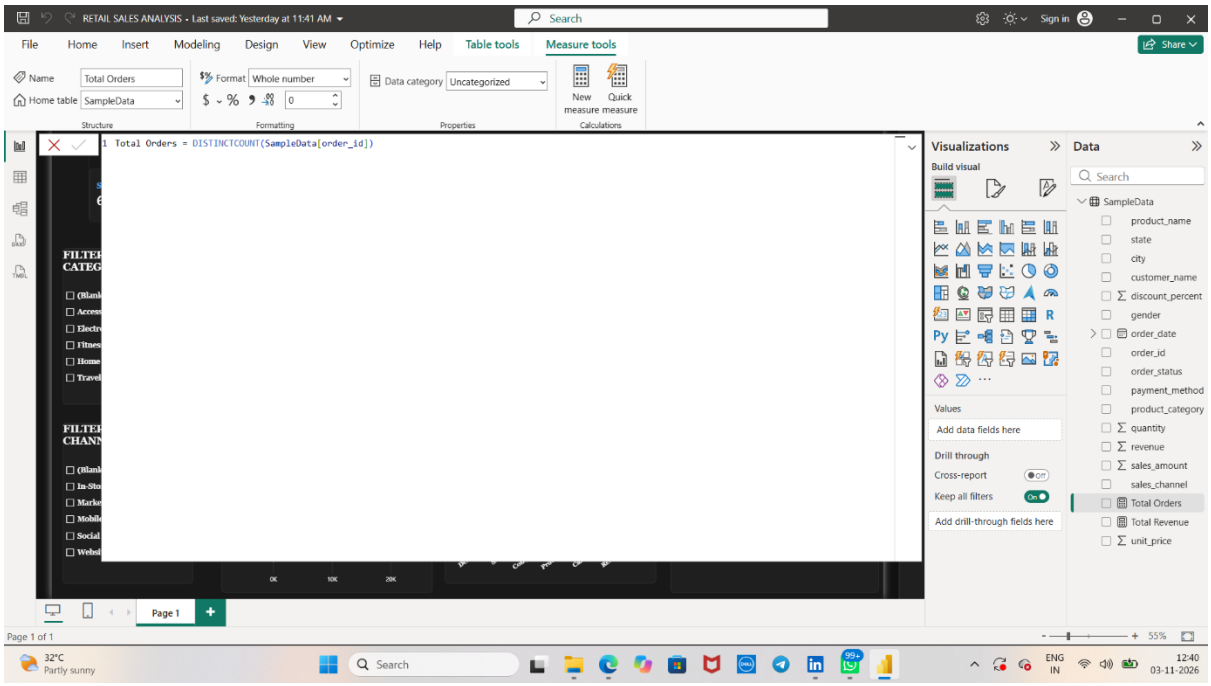


➢ DAX Mesaure:

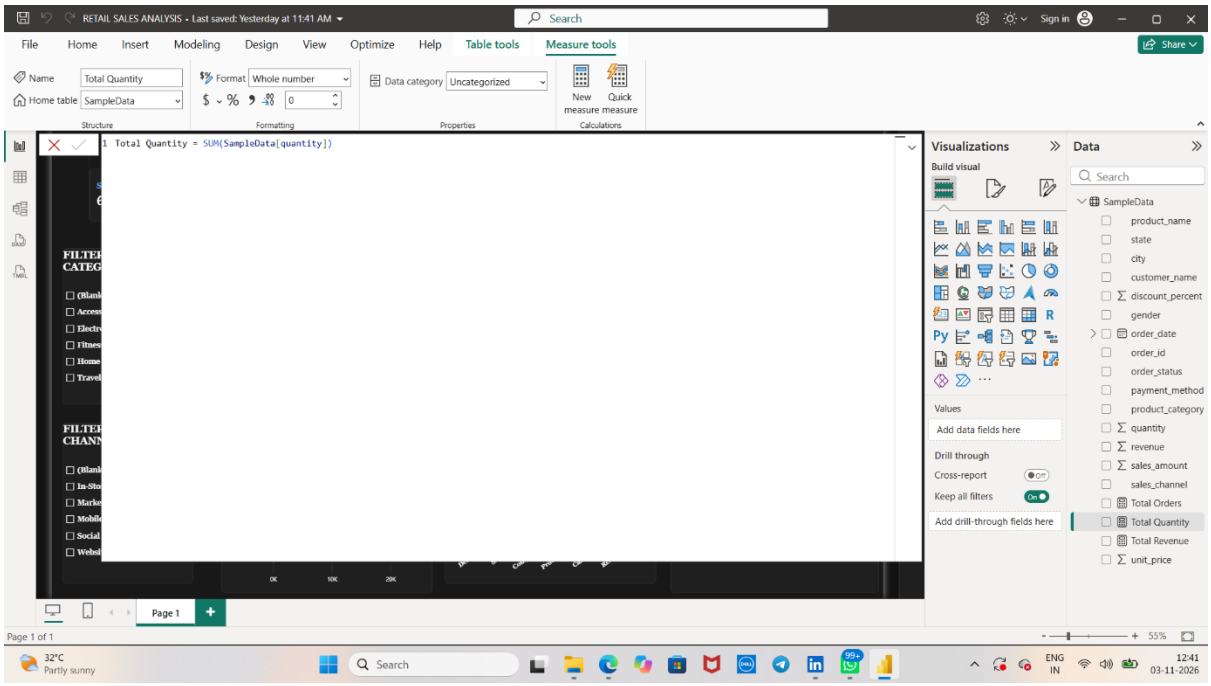
❖ Total Revenue = SUM(SampleData[revenue])



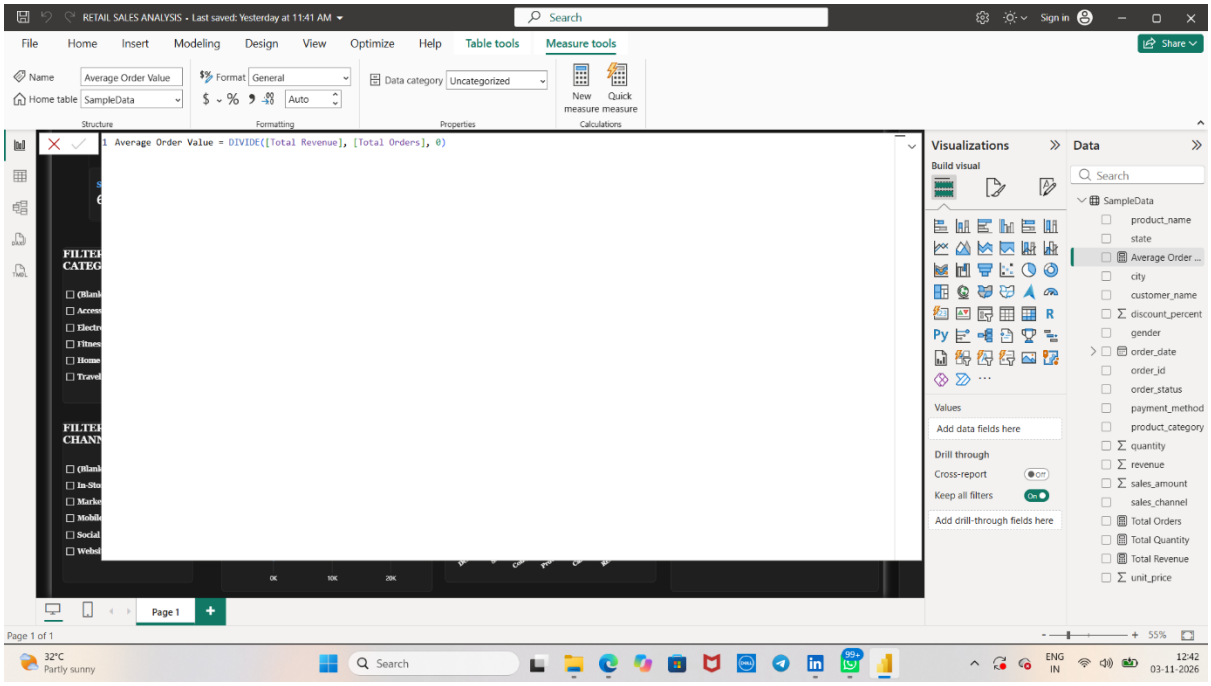
❖ Total Orders = DISTINCTCOUNT(SampleData[order_id])



❖ Total Quantity = SUM(SampleData[quantity])



❖ Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

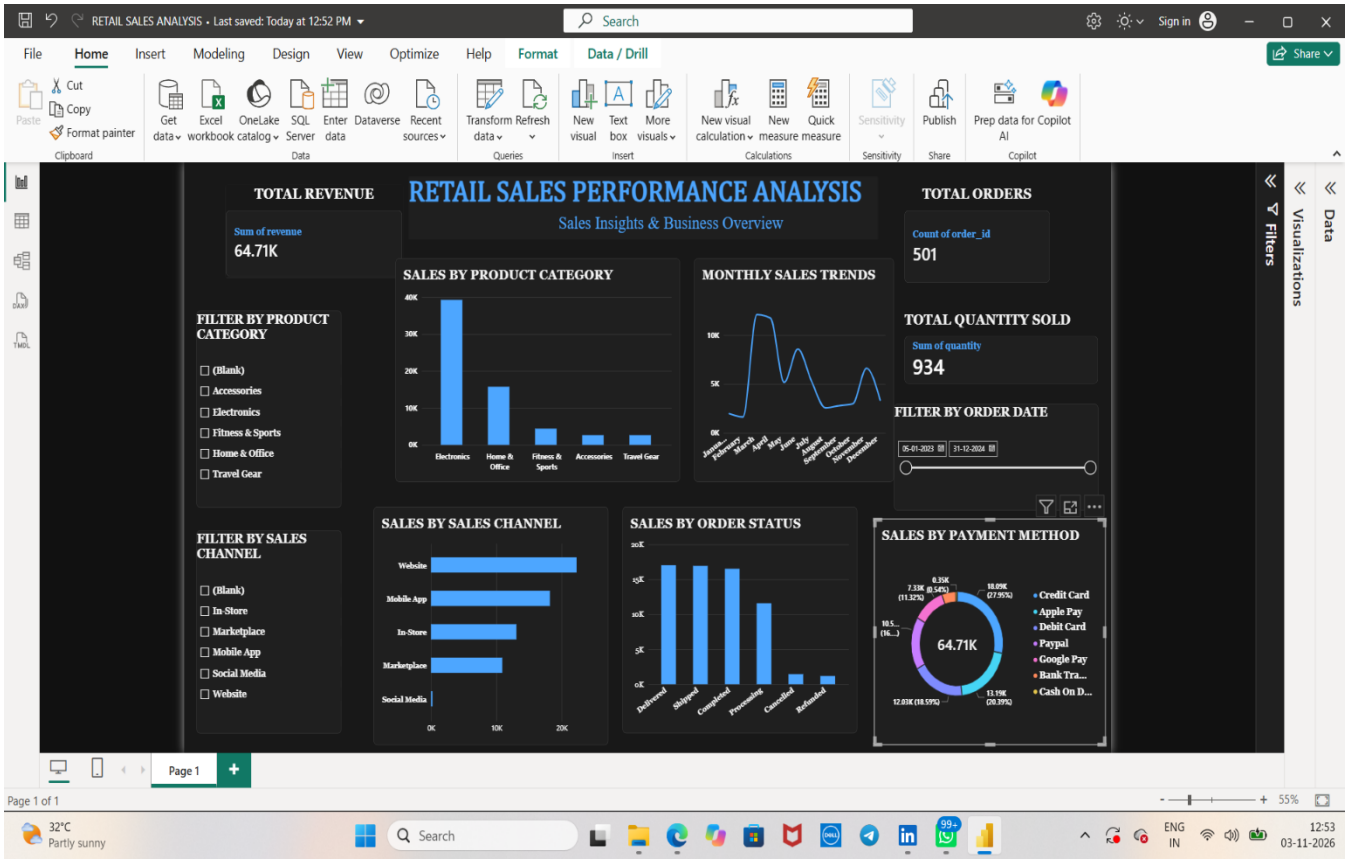


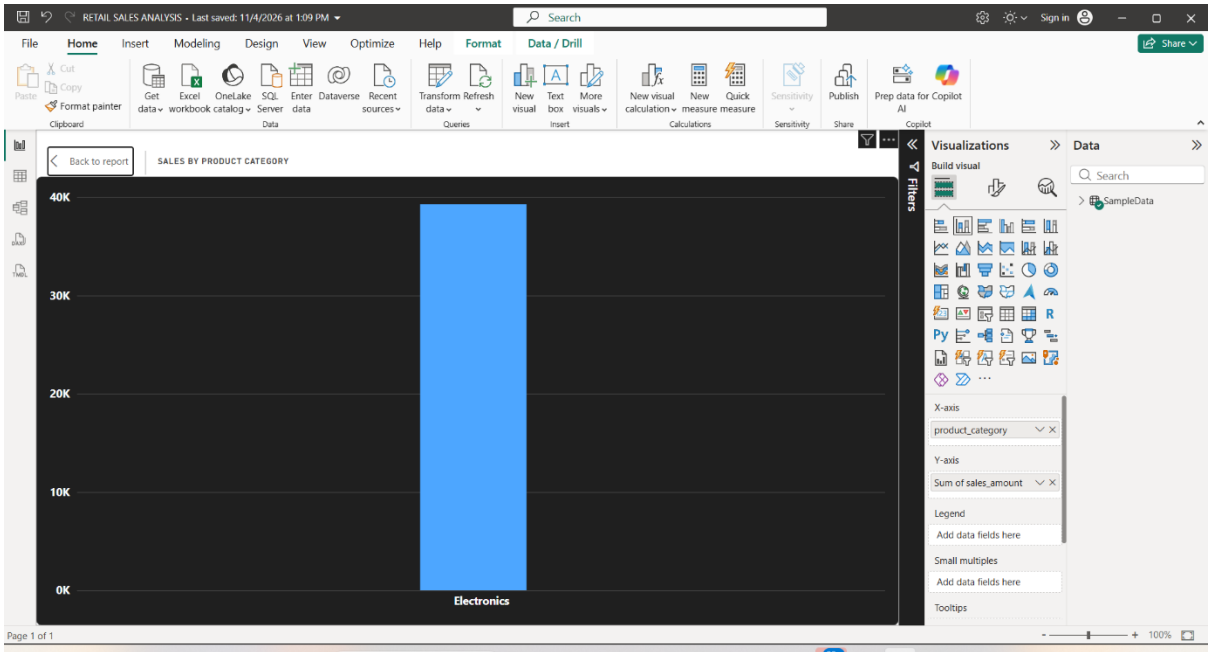
Dax Measure:

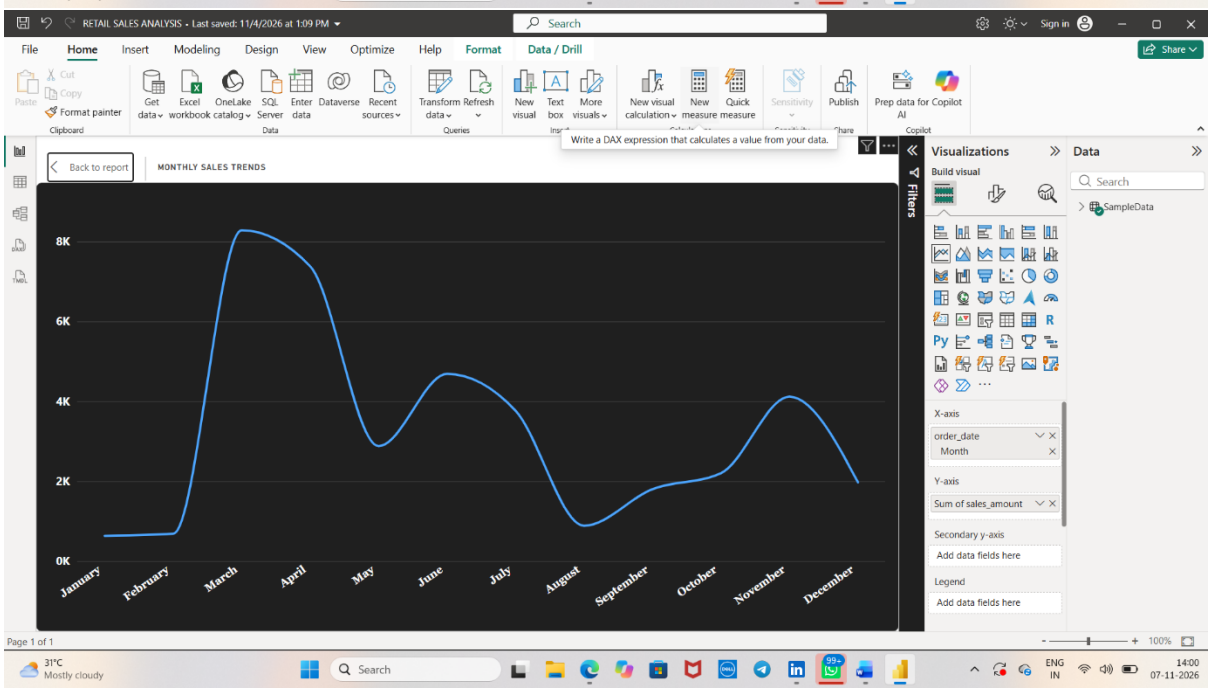
- Total Revenue = SUM(SampleData[revenue])
- DISTINCTCOUNT(SampleData[order_id])
- Total Quantity = SUM(SampleData[quantity])
- Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

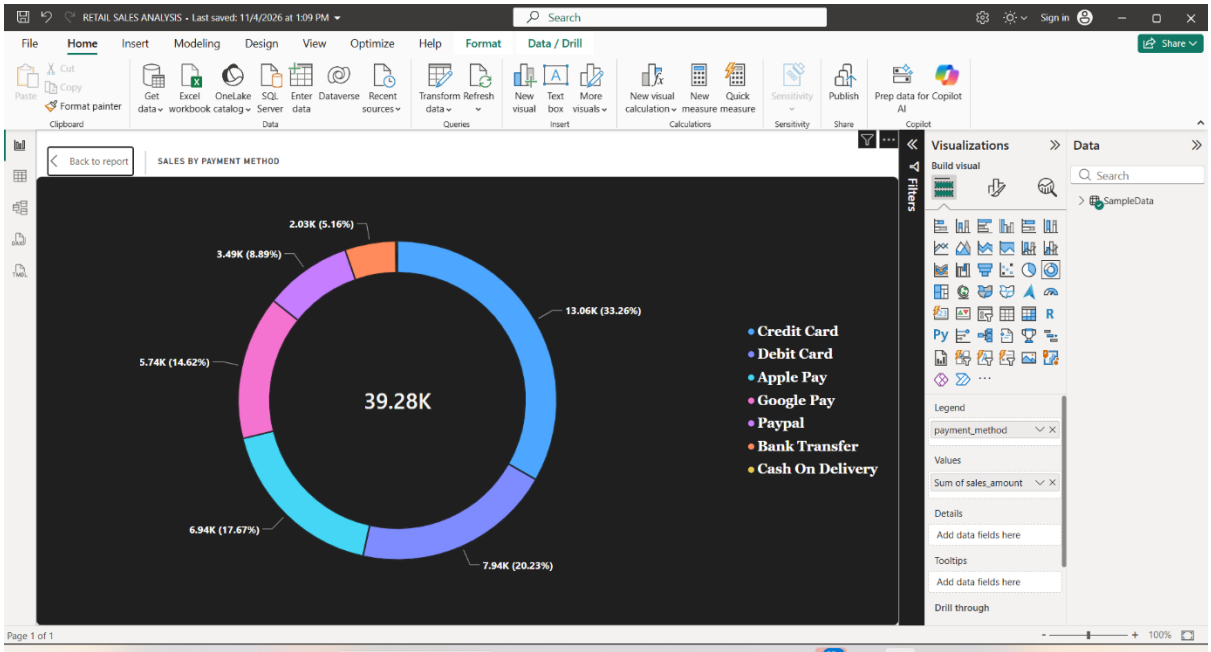
11.Analysis and Visualization:

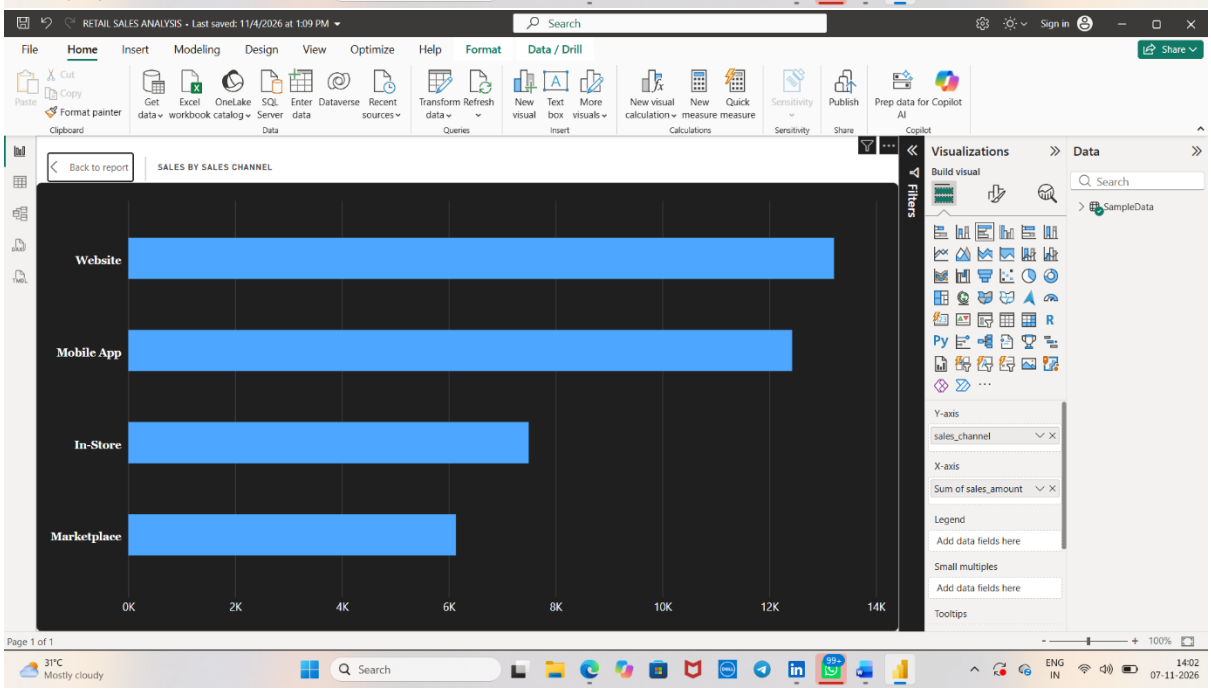
- **DashBoard:**

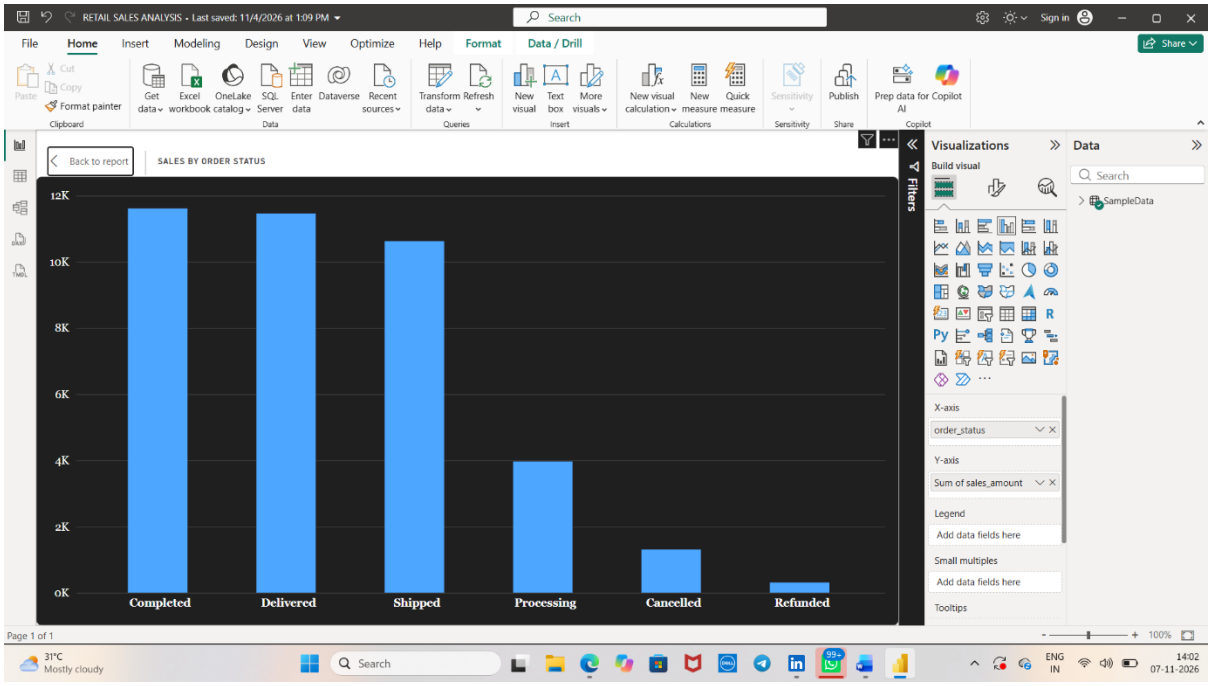

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Excel workbook
OneLake catalog
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Enter data
Dataverse
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Data
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New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
RETAIL SALES PERFORMANCE ANALYSIS
Sales Insights & Business Overview
TOTAL REVENUE
Sum of revenue
64.71K
TOTAL ORDERS
Count of order_id
501
TOTAL QUANTITY SOLD
Sum of quantity
934
SALES BY PRODUCT CATEGORY
Electronics
Home & Office
Fitness & Sports
Accessories
Travel Gear
MONTHLY SALES TRENDS
FILTER BY PRODUCT CATEGORY
(Blank)
Accessories
Electronics
Fitness & Sports
Home & Office
Travel Gear
FILTER BY ORDER DATE
05-01-2023
31-12-2024
FILTER BY SALES CHANNEL
(Blank)
In-Store
Marketplace
Mobile App
Social Media
Website
SALES BY SALES CHANNEL
Website
Mobile App
In-Store
Marketplace
Social Media
SALES BY ORDER STATUS
Delivered
Shipped
Completed
Processing
Cancelled
Refunded
SALES BY PAYMENT METHOD
64.71K
Credit Card
Apple Pay
Debit Card
Paypal
Google Pay
Bank Tra...
Cash On D...
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SALES BY PRODUCT CATEGORY
40K
30K
20K
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Y-axis
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Page 1 of 1
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MONTHLY SALES TRENDS
8K
6K
4K
2K
0K
January
February
March
April
May
June
July
August
September
October
November
December
Visualizations
Build visual
Data
Search
SampleData
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X-axis
order_date
Month
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Sum of sales_amount
Secondary y-axis
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Legend
Add data fields here
Page 1 of 1
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Mostly cloudy
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IN
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07-11-2026


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SALES BY PAYMENT METHOD
2.03K (5.16%)
3.49K (8.89%)
13.06K (33.26%)
5.74K (14.62%)
39.28K
6.94K (17.67%)
7.94K (20.23%)
Credit Card
Debit Card
Apple Pay
Google Pay
Paypal
Bank Transfer
Cash On Delivery
Visualizations
Data
Page 1 of 1


RETAIL SALES ANALYSIS - Last saved: 11/4/2026 at 1:09 PM
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SALES BY SALES CHANNEL
Website
Mobile App
In-Store
Marketplace
0K
2K
4K
6K
8K
10K
12K
14K
Visualizations
Data
Page 1 of 1



➢ Dashboard Features:

❖ The Power BI dashboard was developed to provide a clear and interactive overview of retail sales performance.

➢ Key Performance Indicators (KPIs):

❖ Total Revenue – Displays the overall revenue generated.
❖ Total Orders – Shows the total number of unique orders.
❖ Total Quantity Sold – Displays the total units sold.
❖ Average Order Value – Calculates the average revenue per order.

➢ Visualizations:

❖ Sales by Product Category – Compares sales across product categories.
❖ Monthly Sales Trends – Shows monthly sales performance.
❖ Sales by Sales Channel – Compares sales across different channels.

- ❖ Sales by Order Status – Analyzes sales based on order status.
- ❖ Sales by Payment Method – Shows the contribution of different payment methods.

- ➢ **Interactive Filters:**

- ❖ Product Category, Order Date, and Sales Channel slicers were added to filter the dashboard dynamically.

12.Insights:

Descriptive Analysis:

- ➢ The descriptive analysis summarizes the historical retail sales data using Pivot Tables and Power BI visualizations.

- ❖ **Higher Sales:**

Electronics recorded the highest sales among product categories. Website generated the highest sales among sales channels.

- ❖ **Lower Sales:**

Travel Gear recorded the lowest sales among product categories. Social Media generated the lowest sales among sales channels.

- ➢ The analysis helps understand overall revenue, order counts, product quantities, and sales distribution.

Diagnostic Analysis:

- ➢ The diagnostic analysis compares sales performance across different categories, channels, payment methods, and order statuses.

- ❖ **Higher Sales:**

Electronics had the highest sales amount of 39,277.50. Website recorded the highest sales among channels with 22,282.44.

- **Lower Sales:**

Travel Gear recorded the lowest category sales of 2,642.66. Social Media recorded the lowest channel sales of 241.40.

- These comparisons help identify the stronger and weaker areas of retail sales performance. The data shows the differences but does not establish the reasons behind them.

Predictive Analysis:

- Historical monthly sales trends can be used to understand possible future sales patterns.

- **Higher Sales:**

Electronics and Website sales have shown higher contributions in the analyzed dataset.

- **Lower Sales:**

Travel Gear and Social Media have shown lower sales contributions.

- These historical patterns may help businesses plan future sales activities. However, no separate forecasting model was developed in this project.

Prescriptive Analysis:

- Based on the analysis, the following actions are recommended:

- ✓ Focus on maintaining the availability of high-performing Electronics products.
- ✓ Improve marketing strategies through the Website sales channel.
- ✓ Develop promotional strategies to improve sales of lower-performing categories such as Travel Gear.
- ✓ Explore ways to improve sales through the Social Media channel.
- ✓ Monitor monthly sales trends and customer payment preferences to improve overall sales performance.

13.Conclusions

- ➢ The Retail Sales Performance Analysis project successfully demonstrated the process of cleaning, analyzing, and visualizing retail sales data using Microsoft Excel and Power BI.
- ➢ Excel was used for data cleaning, transformation, formula calculations, and Pivot Table analysis. Power BI was used for data modelling, DAX measures, and creating an interactive dashboard.
- ➢ The project helped identify top-performing product categories, sales channels, payment methods, and order status patterns. The interactive dashboard makes it easier to understand sales performance and supports better business decision-making.
- ➢ Overall, this project provided practical experience in data analysis, visualization, and transforming raw data into meaningful business insights.